Digital signal processing

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1 Conventions

Let $f : \mathbb{R} \to \mathbb{C}$ be a 2π-periodic, integrable function. The Fourier coefficients of f are defined by

$$\hat{f}(k) = \frac{1}{2\pi} \int_0^{2\pi} f(x)\, e^{-ikx}\, \mathrm{d}x, \quad k \in \mathbb{Z}.$$

The Fourier series of f is given by

$$f(x) \;=\; \sum_{k \in \mathbb{Z}} \hat{f}(k)\, e^{ikx}$$

For 2π-periodic functions $f, g \colon \mathbb{R} \to \mathbb{C}$ we define the circular convolution by

$$(f * g)(x) \;=\; \int_0^{2\pi} f(y)\, g(x-y)\, \mathrm{d}y \tag{1}$$

The root of unity:

$$\omega_N = e^{\frac{2\pi i}{N}}$$

2 DFT

Lemma 2.1. *The following holds for roots of unity:*

2.1.1 $\omega_N^{kN} = 1 \; \forall k \in \mathbb{Z}$

2.1.2 N *even.* $\omega_{N/2} = \omega_N^2$

2.1.3 $\overline{\omega_N} = \omega_N^{-1}$

2.1.4 $\omega_N^{kN/2} = (-1)^k \; \forall k \in \mathbb{Z}$

Proof.

1. $$\omega_N^N = \left(e^{\frac{2\pi i}{N}}\right)^N = e^{\frac{2\pi i N}{N}} = e^{2\pi i} = \cos(2\pi) + i\sin(2\pi) = 1$$
$$\omega_N^{kN} = \left(\omega_N^N\right)^k = 1^k = 1$$

2. $\omega_{N/2} = e^{\frac{2\pi i}{N/2}} = e^{2\cdot\frac{2\pi i}{N}} = \left(e^{\frac{2\pi i}{N}}\right)^2 = \omega_N^2$

3. $\overline{\omega_N} = \overline{e^{\frac{2\pi i}{N}}} = e^{-\frac{2\pi i}{N}} = \left(e^{\frac{2\pi i}{N}}\right)^{-1} = \omega_N^{-1}$

4. $\omega_N^{k\cdot N/2} = \left(e^{2\pi i\frac{N/2}{N}}\right)^k = \left(e^{\pi i}\right)^k = (-1)^k$

□

Suppose we have $f : \mathbb{R} \to \mathbb{C}$ which is 2π-periodic. Consider the values of this function at points $\frac{0\cdot 2\pi}{N}$, $\frac{1\cdot 2\pi}{N}$, ..., $\frac{(N-1)\cdot 2\pi}{N}$:

$$x_j := \frac{j\cdot 2\pi}{N}$$

$$y_j := f(x_j) = \sum_{k=-\infty}^{\infty} \hat{f}(k)e^{ikx_j} = \sum_{k=-\infty}^{\infty} \hat{f}(k)e^{2\pi i\frac{kj}{N}} = \sum_{k=-\infty}^{\infty} \hat{f}(k)\omega_N^{kj}$$

By Lemma 2.1.1 we have that $\omega_N^{(k+nN)j} = \omega_N^{kj}\omega_N^{(nj)N} = \omega_N^{kj}$
By grouping the terms with $k_1 = k_2 \mod N$ and obtain the following:

$$y_j = \sum_{k=0}^{N-1} \omega_N^{kj}\left(\sum_{n=-\infty}^{\infty} \hat{f}(k+nN)\right) = \sum_{k=0}^{N-1} \omega_N^{kj} z_k \tag{2}$$

We define the matrix $F_{kj} = \frac{1}{\sqrt{N}}\omega_N^{kj}$. Then:

$$y_j = \sqrt{N}(Fz)_j \Rightarrow y = \sqrt{N}Fz \Rightarrow z = \frac{1}{\sqrt{N}}F^{-1}y = Ty \tag{3}$$

Later we use the vector z to indicate the output of the DFT, namely $F^{-1}y$

Theorem 2.1. *The matrix $F_{kj} = \frac{1}{\sqrt{N}}\omega_N^{kj}$ is unitary*

Proof.

$$(F\overline{F^T})_{jl} = \frac{1}{\sqrt{N}}\overline{\left(\frac{1}{\sqrt{N}}\right)}\sum_{k=0}^{N-1} F_{jk}(\overline{F^T})_{kl} = \frac{1}{N}\sum_{k=0}^{N-1} F_{jk}\overline{F_{lk}} =$$
$$\frac{1}{N}\sum_{k=0}^{N-1}\omega_N^{jk}\overline{\omega_N^{lk}} \underset{(a)}{=} \frac{1}{N}\sum_{k=0}^{N-1}\omega_N^{jk}\omega_N^{-lk} = \frac{1}{N}\sum_{k=0}^{N-1}\omega_N^{(j-l)k}$$

In (a) we used the identity $\overline{\omega_N^{lk}} = \omega_N^{-lk}$ from Lemma 2.1.3

1. $j \neq l$. Using the formula for sum of geometric series:

$$\sum_{k=0}^{N-1}\omega_N^{(j-l)k} = \frac{1-\left(\omega_N^{j-l}\right)^N}{1-\omega^{j-l}} = \frac{1-\left(\omega_N^N\right)^{j-l}}{1-\omega^{j-l}} \underset{(a)}{=} \frac{1-1^{j-l}}{1-\omega^{j-l}} = 0$$

   In (a) we used the identity $\omega_N^N = 1$ from Lemma 2.1.1. The condition $j \neq l$ ensures that the denominator is non-zero, so the expression is well-defined.

2. $j = l$. Then:

$$\sum_{k=0}^{N-1}\omega_N^{(j-l)k} = \sum_{k=0}^{N-1} 1 = N \Rightarrow (F\overline{F^T})_{jl} = \frac{1}{N}N = 1$$

The matrix $FF^* = I$, so F unitary and $F^{-1} = \overline{F^T}$
The entries of F^{-1} are given by $F_{ij}^{-1} = (\overline{F_{ji}}) = \frac{1}{\sqrt{N}}\overline{\left(\omega_N^{ji}\right)} = \frac{1}{\sqrt{N}}\omega_N^{-ij}$. $\square$

Using the result of Theorem 2.1 and equations (3), (2) we define the discrete Fourier transform:

$$z_j = (F^{-1}y)_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kj} y_k \tag{DFT}$$

$$y_j = (Fz)_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{kj} z_k \tag{IDFT}$$

Remark 2.1. Now we make two remarks about the DFT structure:

2.1.1 Firstly using DFT of size n we are unable to detect the frequencies above the (n-1)-th one. In fact as the equation (2) suggests, the frequency of the form $k + nN$ with $k \in 0, ..., N-1$ contributes to the entry with index k. However the maximum frequency human ear can process is approximately 20000, so this issue is handled by choosing an appropriate sample rate, that is usually 44100hz, which is double the maximum frequency which can be detected. The exact reason for it such choice is discussed later. The following plot illustrates the issue discussed (so-called aliasing):

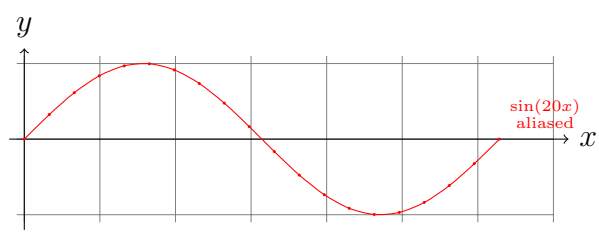

Figure 1: 20 discrete points

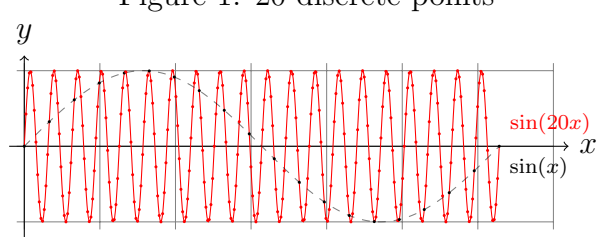

Figure 2: 286 discrete points

2.1.2 The meaning of the phase is often overlooked in the literature, so we consider an example: Let $y = e_k + e_{N-k}$. Then:

$$Ty = T_{*k} + T_{*(N-k)} = \begin{pmatrix} \omega_N^0 \\ \omega_N^{-k} \\ \vdots \\ \omega_N^{-k(N-1)} \end{pmatrix} + \begin{pmatrix} \omega_N^0 \\ \omega_N^{-(N-k)} \\ \vdots \\ \omega_N^{-(N-k)(N-1)} \end{pmatrix}$$

$$\underset{\text{Lemma 2.1.1}}{=} \begin{pmatrix} \omega_N^0 + \omega_N^0 \\ \omega_N^{-k} + \omega_N^k \\ \vdots \\ \omega_N^{-k(N-1)} + \omega_N^{k(N-1)} \end{pmatrix} = \begin{pmatrix} 2\cos(0) \\ 2\cos\left(\frac{2\pi k}{N}\right) \\ \vdots \\ 2\cos\left(\frac{2\pi k(N-1)}{N}\right) \end{pmatrix}$$

As we see from the computation, the output is the function $\cos(kx)$ discretized at points $x_j = \frac{2\pi j}{N}$ for $j \in \{0, \ldots, N-1\}$. What happens if we multiply the positive frequency component of y by ω_N^{-k} and the negative one by ω_N^k?

$$y^* = \omega_N^{-k} e_k + (\omega_N^k e_{N-k})$$

$$Ty^* = \begin{pmatrix} \omega_N^{-k}\omega_N^0 + \omega_N^k\omega_N^0 \\ \omega_N^{-k}\omega_N^{-k} + \omega_N^k\omega_N^k \\ \vdots \\ \omega_N^{-k}\omega_N^{-k(N-1)} + \omega_N^k\omega_N^{k(N-1)} \end{pmatrix}$$

$$= \begin{pmatrix} \omega_N^{-k} + \omega_N^k \\ \omega_N^{-2k} + \omega_N^{2k} \\ \vdots \\ \omega_N^{-kN} + \omega_N^{kN} \end{pmatrix} = \begin{pmatrix} 2\cos\left(\frac{2\pi k}{N}\right) \\ 2\cos\left(\frac{2\pi \cdot 2k}{N}\right) \\ 2\cos\left(\frac{2\pi \cdot 3k}{N}\right) \\ \vdots \\ 2\cos\left(\frac{2\pi kN}{N}\right) \end{pmatrix}$$

We effectively moved the phase of the cosine by $\frac{2\pi}{N}$, since

$$\begin{aligned} (Ty^*)_j &= 2\cos\left(\frac{2\pi k(j+1)}{N}\right) = 2\cos\left(k\left(\frac{2\pi j}{N} + \frac{2\pi}{N}\right)\right) \\ &= 2\cos\left(k\left(x_j + \frac{2\pi}{N}\right)\right) \end{aligned}$$

not hard yet difficult to see by yourself (at least for me)

Theorem 2.2 (Real signal conjugate symmetry)**.** *Let N be even. The following are equivalent:*

1. $y = \overline{y}, y \in \mathbb{C}^N$
2. $z_j = \overline{z_{N-j}} \quad \forall j \in \{1, \dots, N/2 - 1\}$ *and* $z_0, z_{N/2} \in \mathbb{R}$

Proof.
"(1) $\Rightarrow$ (2)" Using the formula for DFT

$$z_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kj} y_k = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \overline{\omega_N^{kj}} y_k \underset{y=\overline{y}}{=} \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \overline{\omega_N^{kj} y_k} =$$

$$\overline{\left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{kj} y_k \right)} \underset{\omega_N^{kN}=1}{=} \overline{\left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kN} \omega_N^{kj} y_k \right)} =$$

$$\overline{\left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-k(N-j)} y_k \right)} = \overline{z_{N-j}}$$

$$z_0 = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-k \cdot 0} y_k = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} y_k \in \mathbb{R}$$

By Lemma 2.1.4:

$$z_{N/2} = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kN/2} y_k = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} (-1)^k y_k \in \mathbb{R}$$

"(2) $\Rightarrow$ (1)" Using the formula for IDFT

$$y_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{kj} z_k = \frac{1}{\sqrt{N}} \left(z_0 + (-1)^j z_{N/2} + \sum_{k=1}^{N/2-1} \omega_N^{kj} z_k + \sum_{l=N/2+1}^{N-1} \omega_N^{lj} z_l \right) \tag{4}$$

$$\sum_{k=1}^{N/2-1} \omega_N^{kj} z_k + \sum_{l=N/2+1}^{N-1} \omega_N^{lj} z_l \underset{l'=N-l}{=} \sum_{k=1}^{N/2-1} \omega_N^{kj} z_k + \sum_{l=1}^{N/2-1} \omega_N^{(N-l)j} z_{N-l} =$$

$$\sum_{k=1}^{N/2-1} \left(\omega_N^{kj} z_k + \omega_N^{(N-k)j} z_{N-k} \right) = \sum_{k=1}^{N/2-1} \left(\omega_N^{kj} z_k + \overline{\omega_N^{kj} z_k} \right) \in \mathbb{R}$$

By plugging this expression back to (4) we get:

$$(4) = \frac{1}{\sqrt{N}} \left(z_0 + (-1)^j z_{N/2} + \sum_{k=1}^{N/2-1} \left(\omega_N^{kj} z_k + \overline{\omega_N^{kj} z_k} \right) \right) \in \mathbb{R}$$

since all the summands and the factor in front are real. $\square$

3 FFT

Theorem 3.1. *Let $y \in \mathbb{C}^N$ and N is even. $K = N/2$. Let:*

$$y^{(1)} = (y_0, y_2, ..., y_{2K-2})^T \quad y^{(2)} = (y_1, y_3, ..., y_{2K-1})^T$$
$$z^{(1)} = F_K^{-1} y^{(1)} \quad z^{(2)} = F_K^{-1} y^{(2)}$$

Then:

$$z_j = \frac{1}{\sqrt{2}} \left(z_j^{(1)} + \omega_N^{-j} z_j^{(2)} \right) \ \forall j \in \{0, \dots, K-1\}$$
$$z_{j+K} = \frac{1}{\sqrt{2}} \left(z_j^{(1)} - \omega_N^{-j} z_j^{(2)} \right) \ \forall j \in \{0, \dots, K-1\}$$

where $z = F_N^{-1} y$

Proof.

$$\begin{aligned} z_j = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{-kj} y_k = \frac{1}{\sqrt{N}} \left(\sum_{\substack{k=0 \\ k \text{ even}}}^{N-1} \omega_N^{-kj} y_k + \sum_{\substack{k=0 \\ k \text{ odd}}}^{N-1} \omega_N^{-kj} y_k \right) = \\ \frac{1}{\sqrt{N}} \left(\sum_{k=0}^{K-1} \omega_N^{-2kj} y_{2k} + \sum_{k=0}^{K-1} \omega_N^{-(2k+1)j} y_{2k+1} \right) \underset{\text{Lemma 2.1.2}}{=} \\ \frac{1}{\sqrt{N}} \left(\sum_{k=0}^{K-1} \omega_K^{-kj} y_{2k} + \omega_N^{-j} \sum_{k=0}^{K-1} \omega_K^{-kj} y_{2k+1} \right) = \\ \frac{1}{\sqrt{2}\sqrt{K}} \left(\sum_{k=0}^{K-1} \omega_K^{-kj} y_{2k} + \omega_N^{-j} \sum_{k=0}^{K-1} \omega_K^{-kj} y_{2k+1} \right) \end{aligned} \tag{5}$$

For $j \in \{0, \dots, K-1\}$ we get the claim immediately. We now focus on the case, when $j = j' + K \in \{K, \dots, N-1\}$. The equation (5) becomes:

$$\begin{aligned} \sqrt{2K} z_j = \sum_{k=0}^{K-1} \omega_K^{-k(j'+K)} y_{2k} + \omega_N^{-(j'+K)} \sum_{k=0}^{K-1} \omega_K^{-k(j'+K)} y_{2k+1} = \\ \sum_{k=0}^{K-1} \omega_K^{-kj'} y_{2k} + \omega_N^{-j'} \omega_N^{-K} \sum_{k=0}^{K-1} \omega_K^{-kj'} y_{2k+1} \underset{\text{Lemma 2.1.4}}{=} \\ \sum_{k=0}^{K-1} \omega_K^{-kj'} y_{2k} - \omega_N^{-j'} \sum_{k=0}^{K-1} \omega_K^{-kj'} y_{2k+1} \end{aligned}$$

which proves the second part of the claim. $\square$

By extending the idea of the idea of the Theorem 3.1 we formulate the following algorithm:

Algorithm 3.1.
Input: $y \in \mathbb{C}^N$, N is a power of two, $n > 1$
Procedure:
if $N > n$:
Get $z^{(1)}$, $z^{(2)}$ by calling Algorithm 3.1 on $y^{(1)}$, $y^{(2)}$. This is well defined since $N/2$ is a power of two

$$z_j = \frac{1}{\sqrt{2}} \left(z_j^{(1)} + \omega_N^{-j} z_j^{(2)} \right) \ \forall j \in \{0, \ldots, K-1\}$$

$$z_{j+K} = \frac{1}{\sqrt{2}} \left(z_j^{(1)} - \omega_N^{-j} z_j^{(2)} \right) \ \forall j \in \{0, \ldots, K-1\}$$

return z
else:
return $F^{-1}y$ (see the expression DFT)

4 Finite impulse response filters

4.1 Convolution theorem

Lemma 4.1. *Let $f : \mathbb{R} \to \mathbb{R}$ have countably many discontinuities. Then f is measurable*

Proof. 1. Let $D \subset \mathbb{R}$ be the set of all discontinuities of f. We have that D has countably many elements. For any set $M \in \mathcal{B}$ we have the following:

$$\begin{aligned} f^{-1}(M) &= \left(f^{-1}(M) \cap D\right) \cup \left(f^{-1}(M) \setminus D\right) \\ &= \left(f^{-1}(M) \cap D\right) \cup \left(\left(f|_{\mathbb{R}\setminus D}\right)^{-1}(M)\right) \end{aligned} \tag{6}$$

We have $(f^{-1}(M) \cap D) \subset D$, so it is countable and hence is Borel measurable, $f|_{\mathbb{R}\setminus D}$ is a measurable mapping since $\mathbb{R} \setminus D$ is a Borel set and the restriction is continuous on its domain. Thus, $f^{-1}(M)$ is the union of two measurable sets, so is measurable as well. That proves measurability of f.

□

Remark 4.1. The statement of the Lemma 4.1 can be relaxed to the set of discontinuities having measure zero. Let F be a generalized distribution function (monotonous and right continuous). Then the following mapping defines an outer measure:

$$\mu^* : \mathcal{P}(\mathbb{R}) \to [0; \infty], A \mapsto \\ \inf\left\{\sum_{n=1}^{\infty} F(b_n) - F(a_n) : A \subset \bigcup_{n=1}^{\infty}(a_n; b_n],\ a_n < b_n\ \forall n \in \mathbb{N}\right\}$$

The Lebesgue measure is $\mu = \mu^*\big|_{\mathcal{B}}$ for $F(x) = x$. However we know that $\mathcal{B} \subset \mathcal{L} := \{A \subset \mathbb{R} : A \text{ is } \mu^* \text{ meaurable}\}$. However we cannot guarantee $\mathcal{B} = \mathcal{L}$ ($\mathcal{L}$ is a sigma algebra by Caratheodori restriction theorem)

Consider a set $N \subset \mathbb{R}$ with $\mu^*(N) = 0$. For any $B \subset \mathbb{R}$:

$$\mu^*(B) = \mu^*((B \cap N) \cup (B \cap N^C)) \leq \mu^*(B \cap N) + \mu^*(B \cap N^C)$$

by sigma subadditivity of μ^*. Moreover by monotonicty of μ^* we have that $\mu^*(B \cap N) \leq \mu^*(N) = 0$, so $\mu^*(B \cap N^C) \geq \mu^*(B)$. By applying monotonicty

again $\mu^*(B \cap N^C) \leq \mu^*(B)$, so $\mu^*(B \cap N^C) = \mu^*(B)$. By combining those results together we obtain $\mu^*(B) = \mu^*(B \cap N) + \mu^*(B \cap N^C)$, so $N \in \mathcal{M}$

Now take a look at (6):

$$f^{-1}(M) = \left(f^{-1}(M) \cap D\right) \cup \left(\left(f|_{\mathbb{R}\setminus D}\right)^{-1}(M)\right)$$

The set $f^{-1}(M) \cap D \subset D$, so it is a null set and hence is in $\mathcal{L}$. Also we know that $\left(f|_{\mathbb{R}\setminus D}\right)^{-1}(M) \subset \mathcal{B} \subset \mathcal{L}$, which implies $f^{-1}(M) \subset \mathcal{L}$. Therefore f is $\mathcal{L} \to \mathcal{B}$ measurable.

Theorem 4.1 (Convolution theorem)**.** *Let $f, g : \mathbb{R} \to \mathbb{R}$ be 2π periodic piecewise continuous and bounded. Then:*

$$\frac{1}{2\pi}\widehat{f * g}(k) = \hat{f}(k)\hat{g}(k)$$

Proof.

$$\begin{aligned} 2\pi \cdot \widehat{f * g}(k) &= \int_0^{2\pi} f * g(x) e^{-ikx} \,\mathrm{d}x \\ &\underset{(1)}{=} \int_0^{2\pi} \left(\int_0^{2\pi} f(y) g(x-y) \,\mathrm{d}y\right) \cdot e^{-ikx} \,\mathrm{d}x \\ &= \int_0^{2\pi} \int_0^{2\pi} f(y) g(x-y) e^{-ikx} \,\mathrm{d}y \,\mathrm{d}x \end{aligned} \tag{7}$$

Let $F(x, y) = f(y)g(x-y)e^{-ikx}$. We have that both f, g are measurabe by Lemma 4.1 since they have finitely many discontinuities. Therefore $\mathrm{Re}(F) = f(\pi_y(x, y))g(x - y)\cos(kx)$ and $\mathrm{Im}(F) = -f(\pi_y(x, y))g(x - y)\sin(kx)$ are $\mathcal{B}_2 \to \mathcal{B}$ measurable.

$$|\mathrm{Re}(F(x, y))|, |\mathrm{Im}(F(x, y))| \leq |F(x, y)| = |f(y)g(x - y)| \leq C < \infty$$

since both f and g are bounded.

$$\begin{aligned} \int_{[0,2\pi]^2} |\mathrm{Re}(F)(x, y)| \,\mathrm{d}\lambda^2(x, y) &\leq \int_{\mathbb{R}^2} C \cdot \mathbb{1}_{[0,2\pi]^2}(x, y) \,\mathrm{d}\lambda^2(x, y) \\ &= C\lambda^2([0, 2\pi]^2) = 4\pi^2 C < \infty \end{aligned}$$

The same holds for $\mathrm{Im}(F)$. Since the function F is absolutely integrable on $[0, 2\pi]^2$ we can use Fubini's theorem. This theorem also ensures, that

$\int_0^{2\pi} f(y)g(x-y)e^{-ikx}\,\mathrm{d}y$ is measurable wherever the integral exist. Moreover, we can say that the integral exists for all y, since the integrand is uniformly bounded.

$$\begin{aligned}\int_0^{2\pi}\int_0^{2\pi} F(x,y)\,\mathrm{d}y\,\mathrm{d}x &= \int_0^{2\pi}\int_0^{2\pi} F(x,y)\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_0^{2\pi}\int_0^{2\pi} f(y)g(x-y)e^{-iky}e^{-ik(x-y)}\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_0^{2\pi} f(y)e^{-iky}\int_0^{2\pi} g(x-y)e^{-ik(x-y)}\,\mathrm{d}x\,\mathrm{d}y\end{aligned}$$

Do the substitution $t = x - y$, $\frac{\partial t}{\partial x} = 1$ (this substitution is a diffeomorphism as a mapping $\Phi : [0; 2\pi] \to [-y, 2\pi - y]$, $x \mapsto x - y$ with $|\det(\mathrm{J}\,\Phi(x))| = 1$)

$$\begin{aligned}\int_0^{2\pi} f(y)e^{-iky}\int_0^{2\pi} g(x-y)e^{-ik(x-y)}\,\mathrm{d}x\,\mathrm{d}y &= \\ \int_0^{2\pi} f(y)e^{-iky}\int_{-y}^{2\pi-y} g(t)e^{-ikt}\,\mathrm{d}t\,\mathrm{d}y &= \\ \int_0^{2\pi} f(y)e^{-iky}\int_0^{2\pi} g(t)e^{-ikt}\,\mathrm{d}t\,\mathrm{d}y &= \\ 2\pi\int_0^{2\pi} f(y)e^{-iky}\hat{g}(k)\,\mathrm{d}y = 4\pi^2\hat{g}(k)\hat{f}(k)\end{aligned}$$

Thus:

$$2\pi\widehat{f*g}(k) = 4\pi^2\hat{g}(k)\hat{f}(k) \Rightarrow \frac{1}{2\pi}\widehat{f*g}(k) = \hat{g}(k)\hat{f}(k)$$

□

Theorem 4.2 (Convolution theorem)**.** *Let $x, y \in \mathbb{R}^n$. Then the following holds:*

$$\frac{1}{\sqrt{N}}(F(x \circledast y)) = Fx \odot Fy$$

where $\circledast$ denotes the circular convolution, and $\odot$ is the component-wise multiplication

Proof. This is the direct analogy to the continuous case in Theorem 4.1 □

4.2 Ideal LPF

Using this knowledge we can derive an ideal low pass filter.

$$z_i = \begin{cases} 0, \ i \in (k; n-k) \\ 1, \ \text{otherwise} \end{cases}$$

This is an impulse reponse of the filter cutting all the frequencies above the k-th one. Recall that the $N-k$ element in DFT vector corresponds to the negative oscilation with the frequency k, due to cyclicity of $e^{\frac{2\pi ijx}{N}}$.

Using the formula for IDFT:

$$y_j = \sum_{l=0}^{N-1} F_{jl} z_k = \sum_{l=0}^{k} F_{jl} + \sum_{l=N-k}^{N-1} F_{jl} = \frac{1}{\sqrt{N}} \left(\sum_{l=0}^{k} \omega_N^{jl} + \sum_{l=N-k}^{N-1} \omega_N^{jl} \right)$$

Substitute $l = N - m - 1$ in the second sum:

$$\frac{1}{\sqrt{N}} \left(\sum_{l=0}^{k} \omega_N^{jl} + \sum_{l=N-k}^{N-1} \omega_N^{jl} \right) = \frac{1}{\sqrt{N}} \left(\sum_{l=0}^{k} \omega_N^{jl} + \sum_{m=0}^{k-1} \omega_N^{j(N-m-1)} \right) =$$

$$\frac{1}{\sqrt{N}} \left(\sum_{l=0}^{k} \omega_N^{jl} + \sum_{m=0}^{k-1} \omega_N^{j(N-m-1)} \right) = \frac{1}{\sqrt{N}} \left(\sum_{l=0}^{k} \omega_N^{jl} + \sum_{m=0}^{k-1} \omega_N^{jN} \omega_N^{-j} \omega_N^{-jm} \right) \underset{2.1.1}{=}$$

$$\frac{1}{\sqrt{N}} \left(\omega_N^{jk} + \sum_{l=0}^{k-1} \omega_N^{jl} + \omega_N^{-j} \sum_{m=0}^{k-1} \omega_N^{-jm} \right)$$

Let $j \neq 0$. Using the sum of geometric series:

$$\sum_{l=0}^{k-1} \omega_N^{jl} = \sum_{l=0}^{k-1} \left(\omega_N^{j}\right)^l = \frac{1 - \left(\omega_N^{j}\right)^k}{1 - \omega_N^{j}}$$

Similarly:

$$\omega_N^{-j} \sum_{m=0}^{k-1} \omega_N^{-jm} = \omega_N^{-j} \sum_{m=0}^{k-1} \left(\omega_N^{-j}\right)^m = \frac{\omega_N^{-j} - \omega_N^{-j} \left(\omega_N^{-j}\right)^k}{1 - \omega_N^{-j}} =$$

$$\frac{\omega_N^{-j} - \omega_N^{-j} \left(\omega_N^{-j}\right)^k}{1 - \omega_N^{-j}} \frac{\omega_N^{j}}{\omega_N^{j}} = \frac{1 - \left(\omega_N^{-j}\right)^k}{\omega_N^{j} - 1} = \frac{\left(\omega_N^{-j}\right)^k - 1}{1 - \omega_N^{j}}$$

By combining the terms together:

$$\sum_{l=0}^{k-1} \omega_N^{jl} + \omega_N^{-j} \sum_{l=0}^{k-1} \omega_N^{-jl} = \frac{1 - \left(\omega_N^j\right)^k}{1 - \omega_N^j} + \frac{\left(\omega_N^{-j}\right)^k - 1}{1 - \omega_N^j} = \frac{\omega^{-jk} - \omega^{jk}}{1 - \omega_N^j}$$

This yields the following expression for y_j, when $j \neq 0$:

$$y_j = \frac{1}{\sqrt{N}} \left(\omega_N^{jk} + \frac{\omega_N^{-jk} - \omega_N^{jk}}{1 - \omega_N^j} \right)$$

If $j = 0$, then all the summands are equal to one, so get just $(2k+1)/\sqrt{N}$. By the Theorem 4.2:

$$\frac{1}{\sqrt{N}}(F(x \circledast y)) = Fx \odot Fy$$

which allows us to implement the low pass filter using its impulse response. That is the point, where theory meets the real world example and makes use of the abstract theorem. In general this concept can be generalaized to any signal, which can be used, for instance for reverb implementation if we have an impulse response of a room (a recording of echo induced by a short impact).

Now we take a closer look on the expression we had derived above:

$$e^{-ix} - e^{ix} = 2\,\mathrm{Im}(e^{-ix}) = 2i\sin(-x) = -2i\sin(x)$$

$$1 - e^{ix} = e^{\frac{ix}{2}} \left(e^{-\frac{ix}{2}} - e^{\frac{ix}{2}} \right) = -2i \cdot e^{\frac{ix}{2}} \sin\left(\frac{x}{2}\right)$$

Therefore:

$$1 - \omega_N^j = 1 - e^{2\pi i \frac{j}{N}} = -2i \cdot e^{\frac{\pi i \cdot j}{N}} \sin\left(\frac{\pi j}{N}\right)$$

$$\omega_N^{-jk} - \omega_N^{jk} = e^{-2\pi i \frac{kj}{N}} - e^{2\pi i \frac{kj}{N}} = -2i \sin\left(\frac{2\pi kj}{N}\right)$$

$$\frac{\omega_N^{-jk} - \omega_N^{jk}}{1 - \omega_N^j} = \frac{-2i \sin\left(\frac{2\pi kj}{N}\right)}{-2i \cdot e^{\frac{\pi i \cdot j}{N}} \sin\left(\frac{\pi j}{N}\right)} = \frac{e^{-\frac{\pi i \cdot j}{N}} \sin\left(\frac{2\pi kj}{N}\right)}{\sin\left(\frac{\pi j}{N}\right)}$$

With $x_j = 2\pi j/N$ we have:

$$\frac{e^{-\frac{\pi i \cdot j}{N}} \sin\left(\frac{2\pi k j}{N}\right)}{\sin\left(\frac{\pi j}{N}\right)} = \frac{e^{-ix_j/2} \sin\left(kx_j\right)}{\sin\left(x_j/2\right)} = \frac{\cos(x_j/2)\sin(kx_j)}{\sin(x_j/2)} - i\sin\left(kx_j\right)$$

Add the term ω_N^{jk} we had previously:

$$\begin{aligned} \omega_N^{jk} + \frac{\cos(x_j/2)\sin(kx_j)}{\sin(x_j/2)} - i\sin(kx_j) = \\ \cos(kx_j) + i\sin(kx_j) + \frac{\cos(x_j/2)\sin(kx_j)}{\sin(x_j/2)} - i\sin(kx_j) = \\ \cos(kx_j) + \frac{\cos(x_j/2)\sin(kx_j)}{\sin(x_j/2)} = \\ \frac{\cos(kx_j)\sin(x_j/2) + \cos(x_j/2)\sin(kx_j)}{\sin(x_j/2)} \end{aligned} \tag{8}$$

Using the equality $\sin(x+y) = \sin(x)\cos(y) + \sin(y)\cos(x)$:

$$(8) = \frac{\sin((k+1/2)x_j)}{\sin(x_j/2)}$$

$$y_j = \begin{cases} \frac{1}{\sqrt{N}} \frac{\sin((k+1/2)x_j)}{\sin(x_j/2)}, \ j \in \{1, \dots, N-1\} \\ (2k+1)/\sqrt{N}, \ j = 0 \end{cases}$$

Miraculously we got a continuous Dirichlet kernel, discretized at points x_j. One can show, that the impulse response of Dirichlet kernel in continuous space acts in the exact same way, namely cuts the frequencies above the k-th one. As we had seen in equation (2), $z_k = \sum_{n\in\mathbb{Z}} \hat{f}(k+nN)$, so given that $\widehat{D_n}(k) = \begin{cases} 1, k \leq n \\ 0, \text{ otherwise} \end{cases}$, this result was not completely unexpected.

5 Infinite impulse response filters

We kick off with the example

Example 5.1. Consider the sequence $x[n] \in \mathbb{R}$, which represents the signal evolving in time. Let:

$$y[0] = x[0], \quad y[n] = x[n] + \alpha y[n-1] \; \forall n > 0$$

The structure of the formula suggests, that we should use an induction to get a closed formula for y:

Base. $y[0] = x[0] = \alpha^0 x[0]$

Step. Assume that it holds for some n, that

$$y[n] = \sum_{j=0}^{n} \alpha^{n-j} x[j] \tag{9}$$

Then:

$$\begin{aligned} y[n+1] = x[n+1] + \alpha y[n] &= \alpha^0 x[n+1] + \alpha \sum_{j=0}^{n} \alpha^{n-j} x[j] \\ &= \alpha^0 x[n+1] + \sum_{j=0}^{n} \alpha^{n+1-j} x[j] = \sum_{j=0}^{n+1} \alpha^{n+1-j} x[j] \end{aligned}$$

Note that the formula (9) can be written as a following convolution:

$$y[n] = \sum_{j=0}^{n} \alpha^{n-j} x[j] = \sum_{j=0}^{n} a_n[n-j] x[j] = x * a_n$$

Thus:

$$a_n = \left(\alpha^0, \alpha^1, \ldots, \alpha^n\right)^T$$

As in the case of Ideal low pass filter in Section 4.2 we got an impulse response of the filter introduced above. The only difference is that this filter has an impulse response with the size proportional to the sample size. That said the IR size grows infinitely in time unlike the IR of an ideal LPF, which has the same size for each frame.

We make an assumption on x, that is absolutely convergent. That is

$$\sum_{n=0}^{\infty} |x[n]| < \infty$$

Then for $n > m$ we have the following estimate for $0 < \alpha < 1$:

$$\begin{aligned}
|y[n] - y[m]| &= \left| \sum_{j=0}^{n} \alpha^{n-j} x[j] - \sum_{j=0}^{m} \alpha^{m-j} x[j] \right| \\
&= \left| \sum_{j=0}^{m} (\alpha^{n-j} - \alpha^{m-j}) x[j] + \sum_{j=m+1}^{n} \alpha^{n-j} x[j] \right| \\
&\leq \left| \sum_{j=0}^{m} (\alpha^{n-j} - \alpha^{m-j}) x[j] \right| + \left| \sum_{j=m+1}^{n} \alpha^{n-j} x[j] \right|
\end{aligned}$$

Since the sum $\sum_{n=0}^{\infty} |x[n]|$ converges we have that x_n is the zero sequence, hence for any $\varepsilon > 0$ exists n_0, s.t. $|x_n| < \varepsilon \; \forall n \geq n_0$:

$$\left| \sum_{j=0}^{m} (\alpha^{n-j} - \alpha^{m-j}) x[j] \right| \leq \sum_{j=0}^{m} |\alpha^{n-j} - \alpha^{m-j}| \cdot |x[j]| =$$

$$\sum_{j=0}^{n_0} |\alpha^{n-j} - \alpha^{m-j}| \cdot |x[j]| + \sum_{j=n_0}^{m} |\alpha^{n-j} - \alpha^{m-j}| \cdot |x[j]| \leq$$

$$\sum_{j=0}^{n_0} |\alpha^{n-j} - \alpha^{m-j}| \cdot |x[j]| + \varepsilon \sum_{j=n_0}^{m} |\alpha^{n-j} - \alpha^{m-j}|$$

With the assumption that $n > m$ we have $n - j > m - j$, so by the monotonicty of α^x we have that $\alpha^{n-j} < \alpha^{m-j} \Rightarrow |\alpha^{n-j} - \alpha^{m-j}| = \alpha^{m-j} - \alpha^{n-j} \leq \alpha^{m-j}$. Moreover $\sum_{j=n_0}^{m} \alpha^{m-j} \leq \sum_{j=0}^{\infty} \alpha^j = C \leq \infty$ since $m - j$ stays positive on the entire range of summation. For the same exact reason $|\alpha^{n-j} - \alpha^{m-j}| = \alpha^{m-j} - \alpha^{n-j} \leq \alpha^{m-n_0}$ if $j \leq n_0$. By combining those two estimates together:

$$\sum_{j=0}^{n_0} |\alpha^{n-j} - \alpha^{m-j}| \cdot |x[j]| + \varepsilon \sum_{j=n_0}^{m} |\alpha^{n-j} - \alpha^{m-j}| \leq$$

$$\alpha^{m-n_0} \sum_{j=0}^{n_0} |x[j]| + C\varepsilon = \alpha^{m-n_0} M + C\varepsilon \to 0$$

as we can choose such m, that $\alpha^{m-n_0}M < \varepsilon$, so the whole expression is bounded by $(C+1)\varepsilon$

For the second term we have that:

$$\left|\sum_{j=m+1}^{n} \alpha^{n-j}x[j]\right| \leq \sum_{j=m+1}^{n} |\alpha^{n-j}||x[j]| \leq \sum_{j=m+1}^{n} |x[j]|$$

which can be made arbitrarily small as the tail sum of a convergent sequence.

This proves for that sufficiently large m we can get $|y[n]-y[m]|$ arbitrarily small, so $y[n]$ is a Cauchy sequence and hence converges.

Lemma 5.1. *Let the series* $\sum_{n=0}^{\infty} |x[n]|$, $x[n] \in \mathbb{C}$ *be absolutely convergent. Then* $\sum_{n=0}^{\infty} |x[n]z^n| < \infty$ *for all* z *with* $|z| \leq 1$

Proof. This directly follows from the comparison principle. Define:

$$a_n = |x[n]| \in \mathbb{R} \quad b_n = |x[n]z^n| \in \mathbb{R}$$

We have that the sequence b_n is bounded by a_n since $|b_n| = |x[n]z^n| = |x[n]||z^n| \leq |x[n]| = a_n$, so using the fact that the series $\sum_{n=0}^{\infty} a_n$ is convergent, we conclude that $\sum_{n=0}^{\infty} b_n$ is convergent as well.

By triangle inequality:

$$\left|\sum_{n=0}^{\infty} x[n]z^n\right| \leq \sum_{n=0}^{\infty} |x[n]z^n| < \infty$$

□

References

[1] Elias M. Stein and Rami Shakarchi (2003) *Fourier Analysis: An Introduction*